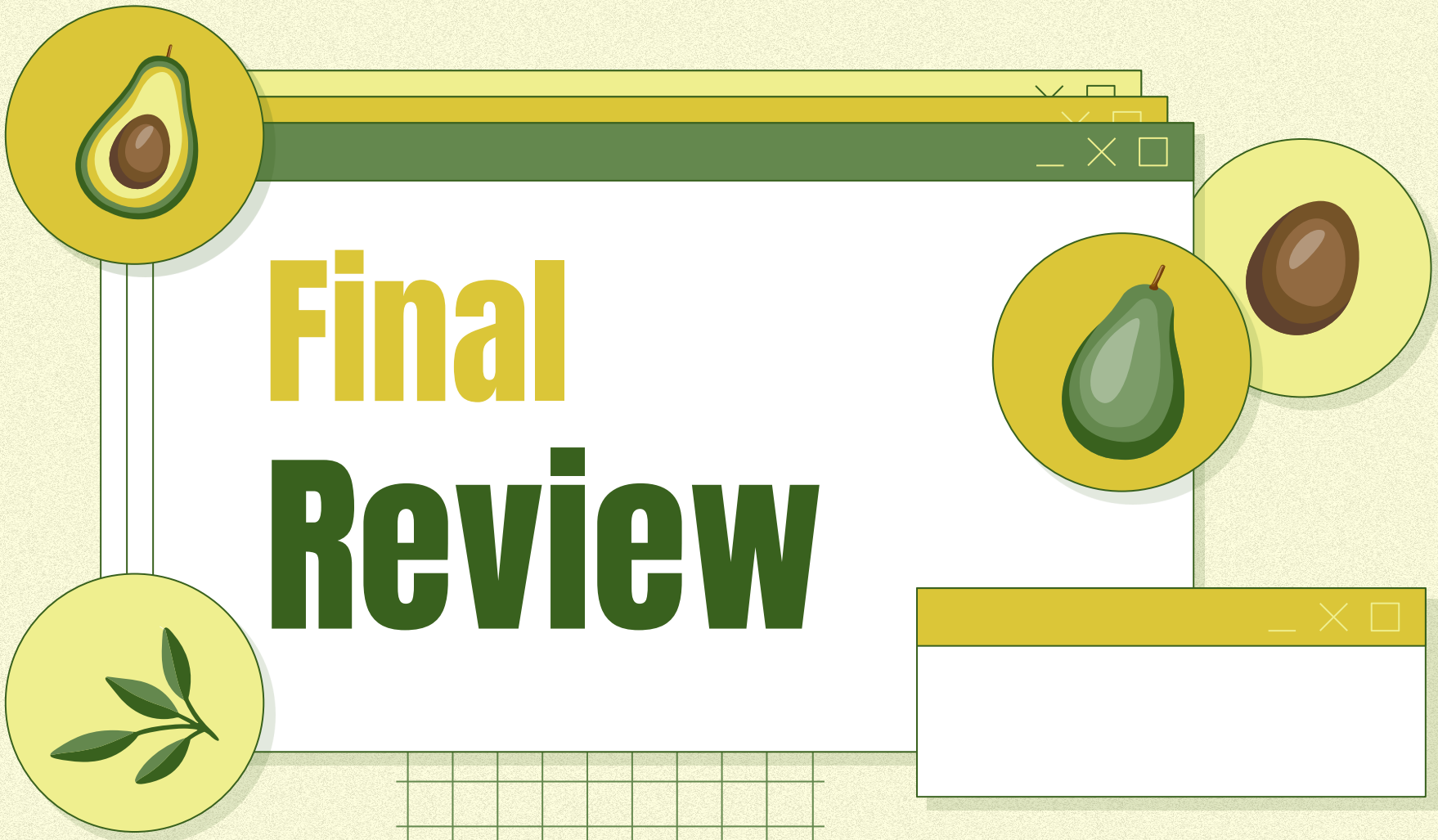




Final Review



Today's AGENDA

Final Logistics

Proctoring guidelines

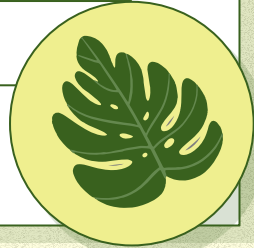
Formatting instructions

Question 1: Multiple Linear Regression and Feature Engineering

Question 2: Combinatorics

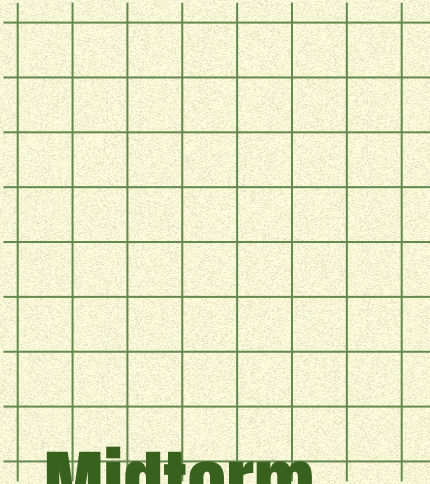
Question 3:

Question 4:



Exam Logistics

Proctoring
Formatting



Midterm Logistics

Format

160 minutes, on paper, no calculators
or electronics except viewing the
exam and proctoring
15 minute submission window (you will
have until 2:35pm PT)

Content

Lectures 1-9, Homeworks 1-8,
Groupworks 1-4

When and Where

Friday, July 31st, 11:30am-2:30pm PT
Zoom

Scanning Test

- 24 pieces of paper → 48 pages
- Compiled into a PDF
- Share to computer
- Upload to gradescope
- Assign pages, one assignment per page of PDF
- 8:58:46

02

Winter 2024 Final Exam Part 1 Problem 4

Multiple Linear
Regression and Feature
Engineering

Albert collected 400 data points from a radiation detector. Each data point contains 3 features: feature A , feature B and feature C . The true particle energy E is also reported. Albert wants to design a linear regression algorithm to predict the energy E of each particle, given a combination of one or more of feature A , B , and C . As the first step, Albert calculated the correlation coefficients among A , B , C and E . He wrote it down in the following table, where each cell of the table represents the correlaton of two terms:

	A	B	C	E
A	1	-0.99	0.13	0.8
B	-0.99	1	0.25	-0.95
C	0.13	0.25	1	0.72
E	0.8	-0.95	0.72	1

Albert further refines his algorithm by fitting a prediction rule of the form:

$$H(A, B, C) = w_0 + w_1 \cdot A \cdot C + w_2 \cdot B^{C-7}$$

Given this prediction rule, what are the dimensions of the design matrix X ?

$$\begin{bmatrix} \end{bmatrix}_{r \times c}$$

So, what are r and c in r rows $\times$ c columns?

Spring 2024

01

Final Exam

Problem 8

Combinatorics

Delta has released a line of trading cards. Each trading card has a picture of a plane and a color. There are 12 plane types, and for each plane type there are three possible colors (gold, silver, and platinum), meaning there are 36 trading cards total.

Now, suppose that each time you board a flight, a pilot hands you 1 card drawn at random from the set of possible cards. You fly 9 times, which gives you **9** cards drawn at random **with replacement** from the set of possible cards.

In this question, you may leave your answers unsimplified, in terms of fractions, exponents, the permutation formula $P(n, k)$, and the binomial coefficient $\binom{n}{k}$.

a

Given that you receive a gold-colored card at least once, what is the probability that you receive a gold-colored card exactly 5 times? Show your work, and put a box around your final answer.

Delta has released a line of trading cards. Each trading card has a picture of a plane and a color. There are 12 plane types, and for each plane type there are three possible colors (gold, silver, and platinum), meaning there are 36 trading cards total.

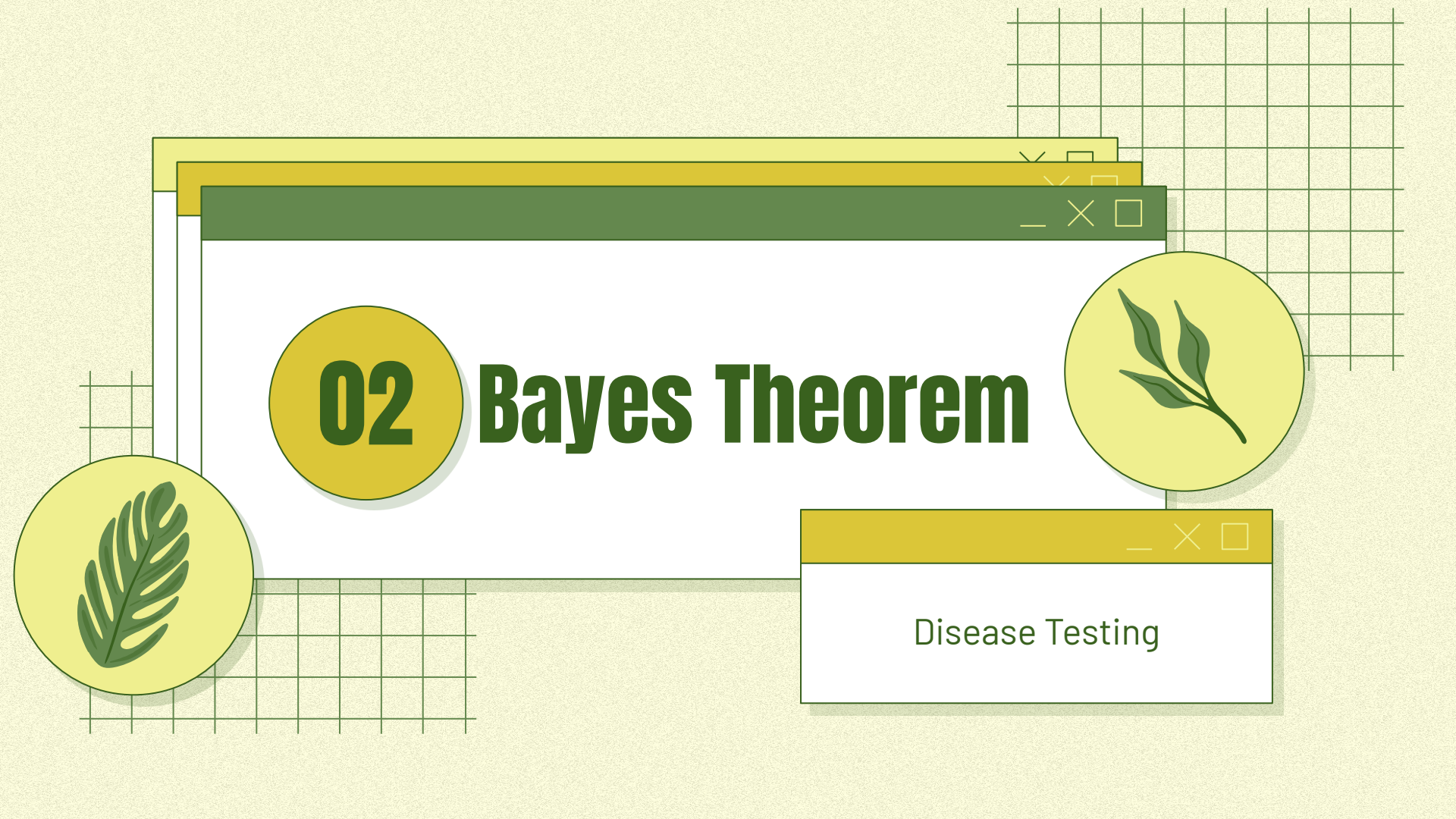
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In this question, you may leave your answers unsimplified, in terms of fractions, exponents, the permutation formula $P(n, k)$, and the binomial coefficient $\binom{n}{k}$.

b

The *Airbus A350*, the *Boeing 767*, and the *Airbus A330* are 3 of the 12 plane types that Delta made trading cards for.

What is the probability that you receive 2 *Airbus A350* cards, 4 *Boeing 767* cards, and 3 *Airbus A330* cards? Show your work, and put a around your final answer.



02

Bayes Theorem

Disease Testing

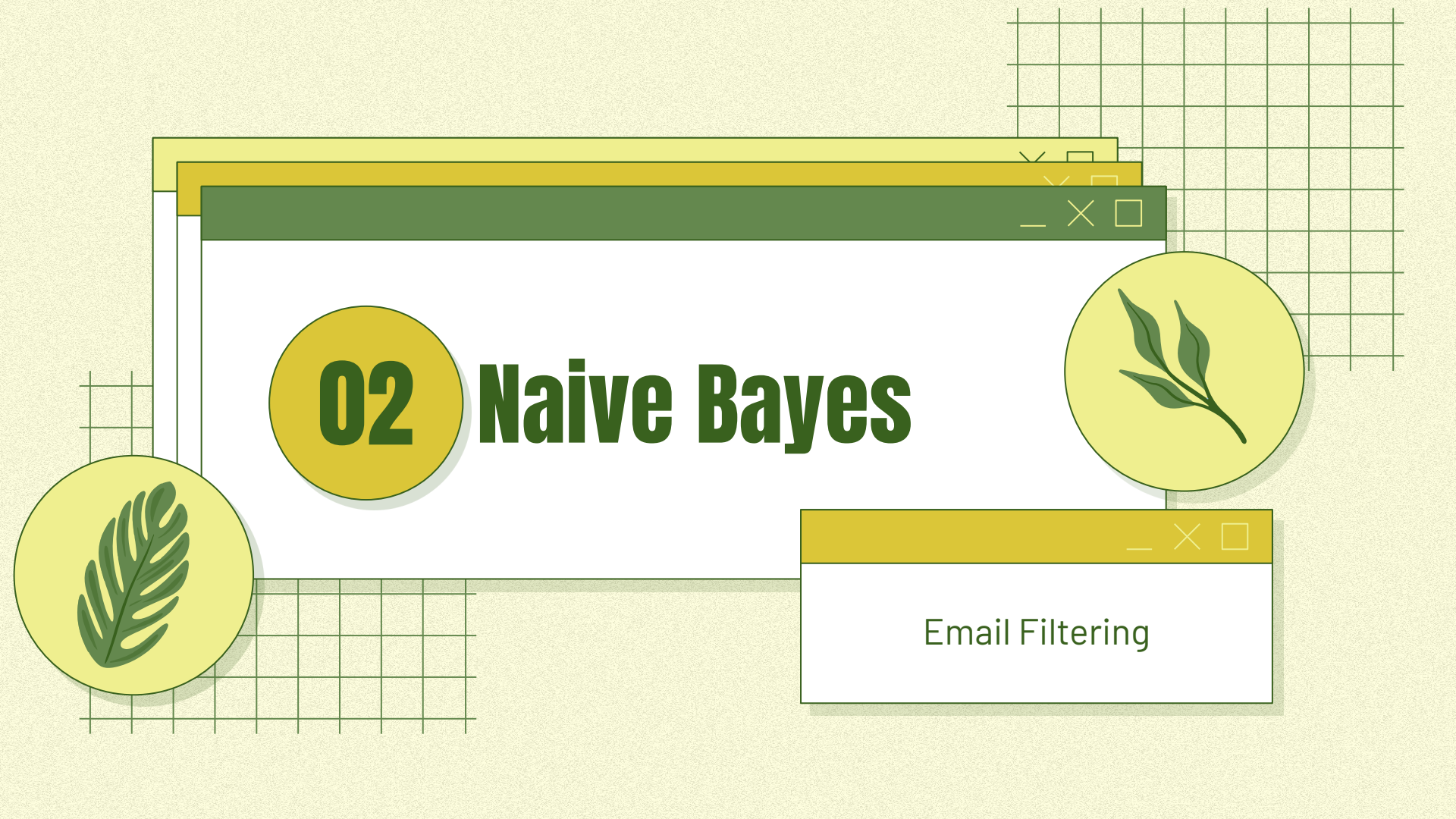
A rare disease affects just 1% of the population. Suppose D is the outcome that a given patient has the disease. The UCSD Hospital has developed a medical test with:

- Sensitivity (true positive rate): $P(+ \mid D) = 0.95$
- Specificity (true negative rate): $P(- \mid \bar{D}) = 0.90$

(a) A random person tests positive. Find $P(D \mid +)$.

(b) Suppose engineers improve the test's specificity to 0.99, with the sensitivity unchanged at 0.95. Recompute $P(D \mid +)$.

(c) Why does $P(D \mid +)$ jump so much in part (b), when sensitivity didn't change at all? What does this reveal about which error type matters most when the base rate is low?



02 Naïve Bayes

Email Filtering

A spam filter uses two features: whether a message contains the word "free" and whether it contains the word "meeting". From training data:

$$P(\text{Spam}) = 0.4, \quad P(\text{Ham}) = 0.6,$$

	Spam	Ham
$P(\text{"free"} \mid \cdot)$	0.8	0.1
$P(\text{"meeting"} \mid \cdot)$	0.05	0.4

(a) A new message contains "free" but not "meeting." Using Naive Bayes, compute the unnormalized scores for Spam and Ham and classify the message. Then give the normalized probability of Spam.



(b) Now suppose the message contains both "free" and "meeting." Classify it.

(c) Why is this method called "naive"? What independence assumption is it making about "free" and "meeting," and can you think of a realistic scenario where that assumption would be violated?

04

Advice and Recommendations

From Jordan and Eric

Advice

Breathe

Studies show that mindfulness can improve exam performance by freeing up brain space from test anxiety

Scavenger Hunt

The exam may or may not be in order of difficulty (which is subjective anyway), so start with what you know

Get Rest

Consistent sleep is an indicator of success.

Explainability

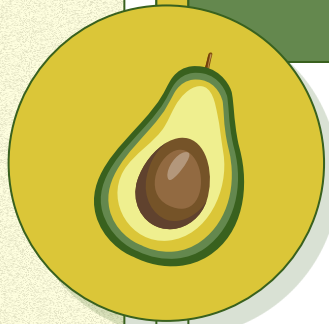
The easiest way to see that you've defended your proof properly is to imagine explaining it to someone with what you've written.

Don't be Hungry

Make sure to eat something before your exam. It'll give a good boost to your blood sugar and keep you energized. In fact, because you have a breakout room to yourself, keep a snack nearby!

Stay Hydrated

Keep a water bottle nearby! You never know when it'll come in handy.



THANKS!

DO YOU HAVE ANY QUESTIONS?

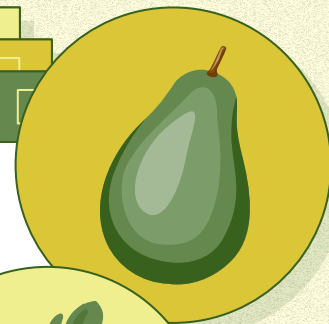
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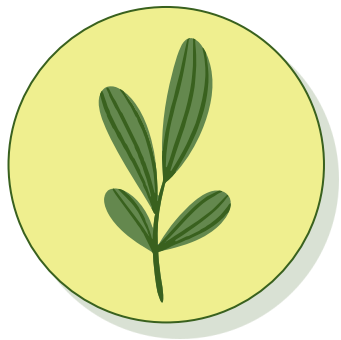
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VECTORS:

- Flat summer pattern design
- Flat tropical summer background with leaves
- Flat summer pattern design (II)



RESOURCES

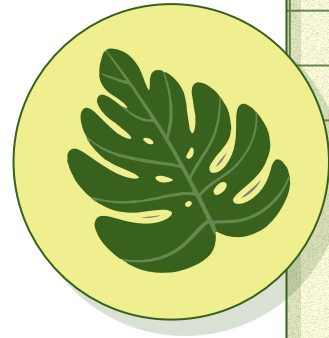
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- Top view of avocado with asparagus and spinach
- Woman standing against wall drinking avocado smoothies in plastic cup
- African american man smiles

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- Flat summer pattern design (II)
- Flat summer pattern design (III)
- Flat summer pattern design (IV)
- Flat tropical summer background with leaves



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